

SYVA: A program to analyze symmetry of molecules based on vector algebra[☆]



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ARTICLE INFO

Article history:

Received 28 August 2016

Received in revised form 15 January 2017

Accepted 20 January 2017

Available online 7 February 2017

Keywords:

Molecular symmetry

Vector algebra

Symmetrization

Framework group

ABSTRACT

Symmetry is a useful concept in physics and chemistry. It can be used to find out some simple properties of a molecule or simplify complex calculations. In this paper a simple vector algebraic method is described to determine all symmetry elements of an arbitrary molecule. To carry out the symmetry analysis, a program has been written, which is also capable of generating the framework group of the molecule, revealing the symmetry properties of normal modes of vibration and symmetrizing the structure. To demonstrate the capabilities of the program, it is compared to other common widely used stand-alone symmetry analyzer (SYMMOL, Symmetrizer) and molecular modeling (NWChem, ORCA, MRCC) software. SYVA can generate input files for molecular modeling programs, e.g. Gaussian, using precisely symmetrized molecular structures. Possible applications are also demonstrated by integrating SYVA with the GAMESS and MRCC software.

Program summary

Program Title: SYVA

Program Files doi: <http://dx.doi.org/10.17632/xd4g8hyjf2.1>

Licensing provisions: GNU GPLv3

Nature of problem: Determination of molecular symmetry elements and symmetry related properties, such as symmetries of vibrational normal modes, framework group, etc.

Solution method: Vector algebraic solution (reflections and rotations) for the generation of symmetry operations, symmetry elements and symmetrization, matrix representation theory for the symmetries of vibrational normal modes.

Unusual features: The program can determine the framework group of molecules, the symmetries of vibrational normal modes and symmetrize the geometry.

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1. Introduction

The word symmetry derives from the Greek words $\sigma\upsilon\nu$ and $\mu\epsilon\tau\rho\omicron\nu$ (meaning with or together and measure). According to American mathematician and physicist Hermann Weyl, something may be considered symmetrical if one can subject it to a certain operation and after the operation it looks the same as it did before [1].

In nature symmetry appears in many forms, therefore its use is very widespread. It can be found both in natural science and art. In everyday language, it suggests some sort of regularity or beauty. This is the reason of its great popularity in various branches of art

in the form of simple geometric shapes or even intricate fractal-like structures [2,3].

In natural sciences, it can be defined as invariance to a certain transformation. It can be found for example in the form of various patterns in biology, in conservation laws of physics, various branches of mathematics and chemistry as well. Its importance is proved by the fact that for the discovery of the mechanism of spontaneous symmetry breaking Japanese-born American scientist Yoichiro Nambu [4] was awarded the Nobel Prize in 2008.

The simplest, original form of symmetry is geometrical symmetry, which means invariance to geometric transformations (rotation, reflection, translation), but it can be defined even on abstract structures, such as self-mappings, automorphism of a graph. The latter with other tools may be used to study certain chemical problems, describe molecules or crystals.

Common areas of its application include molecular spectroscopy and quantum chemistry. It is based on the possibility of

[☆] This paper and its associated computer program are available via the Computer Physics Communication homepage on ScienceDirect (<http://www.sciencedirect.com/science/journal/00104655>).

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mapping the transformation operators of an n -dimensional function space homomorphically to the symmetry operations of an n -atomic molecule, so the transformation operators commute with the Hamiltonian of the molecule. As a result, we can classify the energy eigenvalues and eigenvectors on the basis of the irreducible representations of the point group. With the help of the geometrical symmetry of the molecule, the number of integrals to calculate may be significantly reduced in quantum chemical calculations, based on symmetry, integrals may be identical or disappear [5–8]. An other important area where the exact symmetry of a molecule has to be known is statistical mechanics, namely the rotational partition function contains the molecule's rotational symmetry number [9,10] (the order of the rotational subgroup of its point group). For this reason it is not possible to, for example, evaluate gas phase thermodynamic data or calculate rate coefficients using the transition state theory unless this symmetry number is known. In spectroscopy symmetry can be used to determine some vibrational properties of the molecule, for example the symmetries of normal modes of vibration or the vibrational coordinates. It might also be enough to consider only the symmetry unique atoms to predict IR/Raman properties [10].

Initially the symmetry of a molecule was determined using a flowchart [11–14]. After identification of some symmetry elements, the actual point group can be determined based on the flowchart. However, in case of a larger, more complex molecule it is worth to determine the symmetry elements with a computer. For this purpose there exist matrix algebraic [15–18] as well as graph theoretical [19–23] methods.

The matrix algebraic methods are based on the reflection and rotation matrices representing the symmetry operations or permutation matrices that leave invariant a matrix independent of the frame of reference (e.g. Euclidean distance matrix of the atoms). From these permutations one can generate a permutation group isomorphic to the point group or the complete nuclear permutation-inversion group of the molecule [24–26].

Graph theoretical methods determine the individual symmetry elements using automorphisms of the molecular graph. One of the advantages of these methods is that they do not generate symmetry axes to only a predetermined order, so each of them arises independently of their order. However, their disadvantage is that the permutations of the same nuclei as automorphisms can only be generated combinatorially, therefore the computational cost may be much larger than in case of a matrix algebraic method.

The next section describes how all symmetry elements can be found in a given molecule in a much simpler way, using vector algebra [27]. A Fortran program to carry out the symmetry analysis will be presented as well. The program is also able to determine the framework group, invented by John Pople [28], do some spectroscopic calculations and symmetrize the geometry.

2. Symmetry analysis

The equilibrium molecular geometry [29] can be represented by position vectors of the nuclei. Therefore determining molecular symmetry is possible with simple vector algebraic tools [27,30]. The center of symmetry is represented by a point, an axis by a point and its direction vector while a plane by a point and a normal vector. As the center of mass lies on all symmetry elements, it is invariant to every symmetry operation, so it can be used to describe them.

If we place the center of mass of the molecule into the origin of the coordinate system, the point $\mathbf{0} = (0, 0, 0)$ may be used in each case to describe the symmetry elements. Then the only remaining tasks are the determination of the existence of the center of symmetry, the direction vectors of the rotation axes and the normal vectors of the mirror planes.

In order to determine whether the molecule has a symmetry center, the coordinates of the atoms have to be inverted. If the nuclear configuration after the inversion is identical to the original one, the molecule has a center of symmetry.

To check linearity let us form the difference vectors of the position vectors of the atoms and the center of mass ($\mathbf{P}_i - \mathbf{P}_{\text{com}}$, $i \in \mathbb{Z}_n^+$). The difference vectors express the coordinates of the atoms after moving the center of mass into the origin. The molecule is linear when the difference vectors are all parallel:

$$\left| \frac{(\mathbf{P}_i - \mathbf{P}_{\text{com}}) \cdot (\mathbf{P}_j - \mathbf{P}_{\text{com}})}{\|\mathbf{P}_i - \mathbf{P}_{\text{com}}\| \|\mathbf{P}_j - \mathbf{P}_{\text{com}}\|} \right| = 1, \forall i, j. \quad (1)$$

In this case all the lines defined by these vectors are parallel and they have a common point ($\mathbf{P}_{\text{com}}$), so they coincide. The point group of linear molecules is then readily determined: if the molecule in question has an inversion center, its point group is $D_{\infty h}$, otherwise it is $C_{\infty v}$. All of its symmetry elements can easily be described with the direction vector of the axis of the molecule and perpendicular vectors.

The determination of planarity happens in a similar way. In this case we have to decide whether the atoms constituting the molecule are located on the same plane. This can be done by generating the normal vector ($\mathbf{n}$) of the possible plane as the cross product of any two difference vectors, then calculating the scalar product of all the difference vectors and this normal vector. If all the difference vectors are orthogonal ($(\mathbf{P}_i - \mathbf{P}_{\text{com}}) \cdot \mathbf{n} = 0$, $\forall i$) to the normal vector, the molecule is located in the plane defined by the normal vector and the center of mass.

Let us introduce the projection of the vector $\mathbf{v}$ onto the vector $\mathbf{u}$:

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|^2} \mathbf{u}. \quad (2)$$

With the help of Eq. (2), mirroring to the plane with a normal vector $\mathbf{n}$ and containing the origin can be specified as follows:

$$\mathbf{P}' = \mathbf{P} - 2 \text{proj}_{\mathbf{n}}(\mathbf{P}) \quad (3)$$

where $\mathbf{P}'$ is the image of the point $\mathbf{P}$ after the transformation. A general reflection to a plane can then easily be given using a point of the plane ($\mathbf{P}_0$). For this the origin of the coordinate system is formally placed into the point $\mathbf{P}_0$:

$$\mathbf{P}' = (\mathbf{P} - \mathbf{P}_0)' + \mathbf{P}_0 = \mathbf{P} - 2 \text{proj}_{\mathbf{n}}(\mathbf{P} - \mathbf{P}_0). \quad (4)$$

A symmetry plane whether contain all atoms, in which case its normal vector has already been determined with its planarity, or there are an even number of atoms outside of the plane, which transform into each other pairwise. Normal vectors of further possible symmetry planes are therefore constructed as the difference between position vectors of two nuclei with same atomic numbers ($\mathbf{n} = \mathbf{P}_i - \mathbf{P}_j$). The plane being examined is a symmetry plane, if carrying out the reflection ($\mathbf{P}_0 = \mathbf{0}$ and $\mathbf{n} = \mathbf{P}_i - \mathbf{P}_j$) according to Eq. (4) a structure same as the original one is obtained.

Execution of rotations may happen using Rodrigues' rotation formula [31]:

$$\mathbf{P}' = \mathbf{P} + \sin \varphi \mathbf{d} \times \mathbf{P} + (1 + \cos \varphi) \mathbf{d} \times (\mathbf{d} \times \mathbf{P}), \quad (5)$$

where $\mathbf{P}'$ is the vector obtained by the rotation of an angle φ around the line passing through the origin with a direction vector $\mathbf{d}$. Rotation around a general axis, passing through $\mathbf{P}_0$, can be performed as before by shifting $\mathbf{P}_0$ to the origin.

A direction vector of a symmetry axis may be the axis of the molecule if it is linear, the line perpendicular to the plane of the molecule in case it is planar, or can be constructed from the position vectors of two (C_2 axes) or three (C_n axes, $n > 2$) atoms

with same atomic numbers. Direction vectors of possible C_2 axes can be written using the position vectors of atoms i and j as follows:

$$\mathbf{d} = \frac{\mathbf{P}_i + \mathbf{P}_j}{\|\mathbf{P}_i + \mathbf{P}_j\|}, \quad (6)$$

if $\mathbf{d} \neq \mathbf{0}$. Direction vectors of higher order axes can be given with the position vectors of three atoms (i, j, k) as

$$\mathbf{d} = \frac{(\mathbf{P}_i - \mathbf{P}_j) \times (\mathbf{P}_j - \mathbf{P}_k)}{\|(\mathbf{P}_i - \mathbf{P}_j) \times (\mathbf{P}_j - \mathbf{P}_k)\|}. \quad (7)$$

It is enough to consider the cases in which the atoms i, j and k correspond to adjacent vertices of a regular polygon. Knowing the possible direction vectors of the symmetry axes, we can perform rotations around them by an angle of $\varphi = 2\pi/n$ according to Eq. (5). By systematic variation of the value of n , determination of the axis' order is also possible.

Determination of the improper rotation axes happens similarly to the determination of proper rotation axes, but the rotation is followed by a reflection onto the plane defined by the center of mass and the direction vector of the axis.

After localization of the symmetry elements of the molecule, every symmetry operation can be easily given, and knowing the number of various symmetry operations (i, σ, C_n, S_n) the point group and the exact type of symmetry operations ($C'_2, C''_2, \sigma_v, \sigma_d$) can be determined as well.

To find out the framework group, we have to position the individual atoms in the appropriate symmetry subspaces. The difference of the atomic position vectors in the first subspace (O) and the center of mass disappear, while the position of other atoms can be determined by projections according to Eq. (2). If the atom is situated on a rotation axis, the difference between its position vector ($\mathbf{P}_i$) and its projection onto the direction vector of the axis ($\mathbf{d}$); if it lies on a reflection plane, the projection of its position vector onto the normal vector of the plane ($\mathbf{n}$) disappears:

$$C_n : \mathbf{P}_i - \text{proj}_{\mathbf{d}}(\mathbf{P}_i) = \mathbf{0} \quad (8)$$

$$\sigma : \text{proj}_{\mathbf{n}}(\mathbf{P}_i) = \mathbf{0}. \quad (9)$$

Given the symmetry elements, the atoms can be easily mapped to the symmetry subspaces, and the framework group of the molecule can be written providing more information about its symmetry.

Using the character table of the point group, the symmetry properties of the normal modes of vibration may also be resolved. For this we need to construct a reducible representation on the basis of normal coordinates. In this representation, the characters of the symmetry operations can be formulated as follows:

$$C_n^m : \chi^{\text{red}}(R) = n_R \left(1 + 2 \cos \left(m \frac{2\pi}{n} \right) \right) \quad (10)$$

$$S_n^m : \chi^{\text{red}}(R) = n_R \left(-1 + 2 \cos \left(m \frac{2\pi}{n} \right) \right) \quad (11)$$

$$E = C_1, i = S_2, \sigma = S_1, \quad (12)$$

n_R is the number of atoms invariant to the operation R , these atoms are located on the symmetry element corresponding to R . The number of these atoms had been obtained earlier, when the framework group was determined.

This representation can be reduced using character tables, this way we can specify how many times an irreducible representation of the group (μ) occurs in the previously generated reducible representation:

$$a_\mu = g^{-1} \sum_R \chi^{\text{red}}(R) \chi^\mu(R), \quad (13)$$

where g is the order of the group, $\chi^{\text{red}}(R)$ and $\chi^\mu(R)$ the characters of the symmetry operations R in the reducible and irreducible

representations, respectively. Subtracting the irreducible representations corresponding to the translation and the rotation, we get the symmetry properties of the vibrational normal modes of the molecule.

After determination of the symmetry operations, it is also possible to change (i.e. optimize) the geometry of the molecule according to the whole point group or any of its subgroups. In other words, we can symmetrize the structure by placing the atoms exactly on the symmetry elements. In this case, the symmetry elements need to be fixed first to represent the ideal symmetry. This idealization of the symmetry elements can be made by orthogonalizing or parallelizing them and adjusting the angle between each pair of elements. The adjusting of the angle is made by rotating the direction or normal vector around the principal axis. When the norm of the difference of a normal or direction vector and the rotated vector is small enough, the transformed vector is used instead of the original one. This rotated vector is considered as the ideal vector describing that particular symmetry element. The direction vector of the principal axis is used instead of the vectors parallel with it. This symmetrization can be made using only geometrical considerations, there is no need to compute energies or any energy related quantities like its derivatives. Therefore a perfectly symmetrical nuclear configuration can be produced in a very simple and quick way. This symmetrized structure may be used later in quantum chemical calculations, for example, to reduce the running time of these calculations.

The final step of the symmetrization is the determination of the exact positions of the atoms, appropriate to the chosen symmetry. First, we have to project an atom which has not yet been optimized onto the symmetry element containing it. The new, accurate position vectors of the atoms are as follows:

$$C_n : \text{proj}_{\mathbf{d}}(\mathbf{P}_i) \quad (14)$$

$$\sigma : \mathbf{P}_i - \text{proj}_{\mathbf{n}}(\mathbf{P}_i), \quad (15)$$

$\mathbf{d}$ is the direction vector of the axis C_n , $\mathbf{n}$ is the normal vector of the plane σ and $\mathbf{P}_i$ is the original position vector of the atom i . The last step is the application of all symmetry operations on these new position vectors. After performing an operation, we determine the closest atom to the transformed position vector and replace its position vector with the transformed one.

The previously described procedures can, of course, be performed on a subset of the atoms of the whole molecule as well. This could be useful when the entire molecule is not symmetric – or it has a lower symmetry – but there is a part of it which has a higher symmetry. This is commonly the case in transition metal complexes, for example.

3. Program description

For the symmetry analysis of an arbitrary molecule, we created a program in Fortran. The structure of the program is shown in Fig. 1.

It can be seen that the whole program is based on dot and cross product procedures executing vector algebraic operations.

The program generates all symmetry elements of the molecule (sym_elements subroutine) as described in the previous section. After performing the possible symmetry operations (reflect, rotate, srotate, inversion procedures), check subroutine determines whether the resulting nuclear configuration is the same as the original one. The previous equations are, of course, valid within a certain tolerance represented by the delta distortion parameter. This tolerance is the largest allowed difference of the appropriate transformed coordinates from the original ones. The subroutine also generates values to characterize the degree of distortion. These express the maximum deviation of the symmetry elements – from being parallel or orthogonal to each other – and

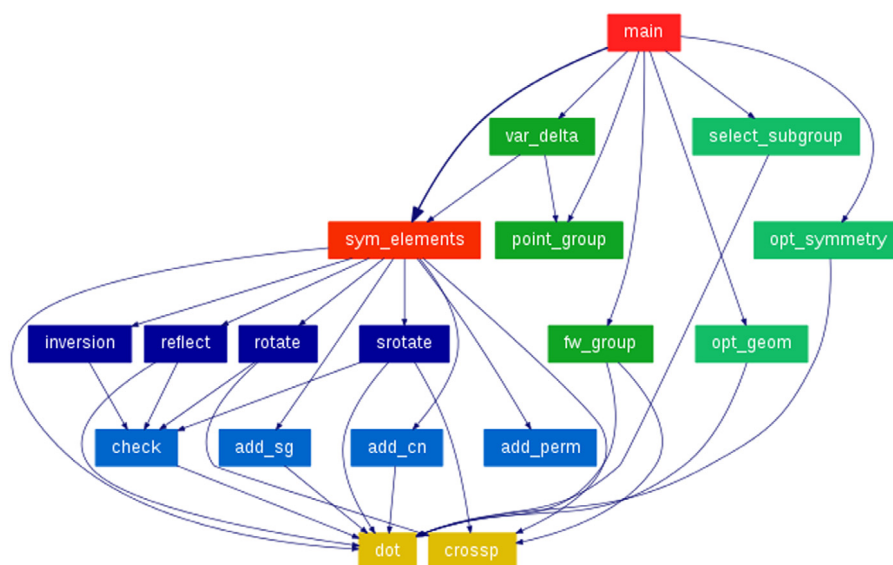


Fig. 1. The structure of the program SYVA.

the position vectors of the atoms from ideal, i.e. being exactly on the symmetry elements.

The subroutine specifies the nuclear permutations according to the symmetry operations, too. The point group of the molecule is decided by the `point_group` subroutine based on the number of symmetry operations, then the `fw_group` subroutine deduces the exact type of symmetry operations and the framework group of the molecule. The tolerance used during the framework group generation is the upper bound to the distance of an atom from the symmetry element it is located on.

Utilizing the character table of the point group, `mkrep` and `re-drep` subroutines define and reduce the representation generated by the normal coordinates, giving the symmetry of normal modes of vibration.

The highest order point group is revealed by the `var_del` subroutine using a higher tolerance (varying the tolerance in an interval from `tol_sup` to `tol_inf`, these bounds can be specified using command line arguments), then `select_subgroup` subroutine picks any or all of the symmetry operations constructing a subgroup. Optimization of the symmetry elements and the geometry is performed by `opt_symmetry` and `opt_geom` subroutines. The symmetrized geometry should show the chosen point group within numerical error.

After the symmetrization it is possible to generate a Gaussian [32] input file containing the symmetrized geometry. This ensures the use of the correct symmetry, lowering the number of degrees of freedom, therefore reducing the computational time. Subsequent revision of the generated input file (level of theory, charge, multiplicity) might be needed. It is straightforward to extend the number of known file formats, if necessary, making the symmetrized geometries available to use with other program packages as well.

4. Application

Let us consider the tetracyclopentylmethane molecule as an example. The molecular structure can be seen in Fig. 2.

Fig. 5. shows the input coordinates and some results of the program. The point group is the same as published by Kozhushkov et al. (S_4) [33]. Among the results the location of the symmetry elements can also be found, and they can be displayed if needed as well. From the framework group, it turns out that a single carbon

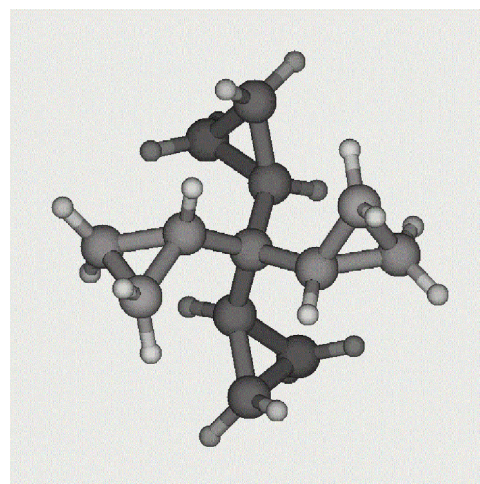


Fig. 2. The equilibrium geometry of tetracyclopentylmethane molecule.

atom is located in the (O) subspace, it is invariant to every symmetry operation. Other atoms do not lie on symmetry elements.

It can be seen that the molecule has 70 different Raman active vibrational frequencies from which 47 are IR active, too. Therefore we can expect the appearance of so many bands in the spectra in case of an accurate, not distorted geometry. Among the results the symmetry properties of the vibrations are also included.

The highest order point group turns out to be S_4 , as expected. It is possible to generate a geometry for further calculations according to the S_4 point group or any of its subgroups (C_1 , C_2) to take into account certain distortions if necessary. It can be seen in Figs. 5 and 6 that the symmetrization was successful. The value representing the distortion of the structure (Distortion of geometry due to symmetry elements) decreased from 0.0529 (obtained with a tolerance of 0.1 Å) to 7×10^{-8} (obtained with the default tolerance, 0.001 Å). This means that the atoms are almost exactly on the symmetry elements after symmetrization. The remaining error relates only to the precision of the output (10^{-8} Å). The S_4 point group could also be found with a much higher tolerance.

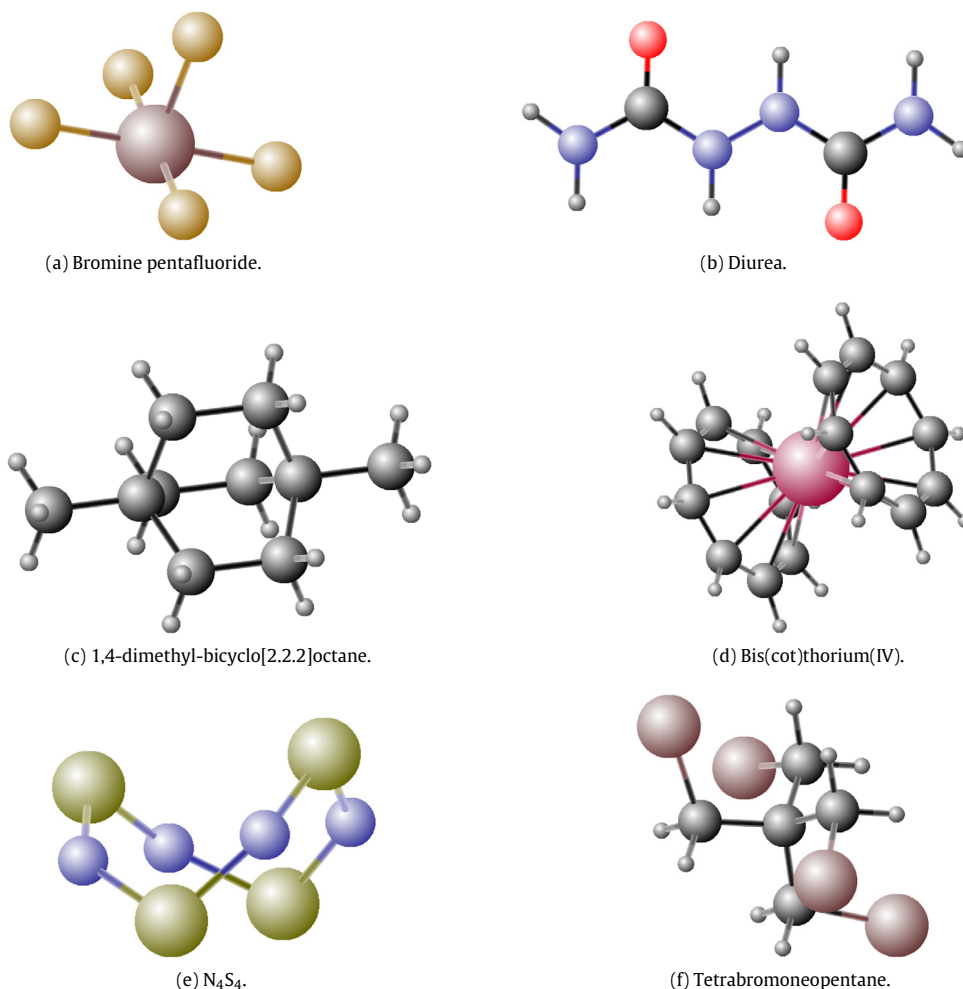


Fig. 3. Initial structures of the molecules listed in Table 3.

Table 1

Average running times on one molecule in milliseconds.

Program	SYVA	SYMMOL	Symmetry	Symmetrizer
Avg. time (ms)	8.8	9.3	51.7	311.8

5. Comparison and integration with other programs

For comparison let us consider some other programs, namely SYMMOL [17,34–36] (the version included in the IQmol software), symmetry [37], written by S. Patchkovskii, which are all licensed under GNU GPL, and Symmetrizer [18] licensed under GNU LGPL. The three programs had been compiled using the same level of optimization, and calculations were performed on 235 different geometries [38,39] with various symmetries. Average running times, some assigned point groups, the structure of the listed molecules and the used input geometries can be seen in Table 1, Table 3, Fig. 3 and Fig. 4, respectively.

As can be seen, Symmetrizer, written in Java, is the slowest, symmetry is the third since it always performs optimization of the symmetry elements, lastly SYVA and SYMMOL have the shortest average running times on the molecules of the test set, SYVA being slightly (about 5%) faster.

In the following let us also consider three other programs, NWChem [40], ORCA [41] and MRCC.[42] Each of these quantum chemical software is capable of utilizing point group symmetry to some extent and are widely used.

In regard to the point groups determined, most of them were correct in case of the stand-alone software. However, with some very distorted geometries, there were about 1, 5, 16 and 26 point groups with SYVA, symmetry, Symmetrizer and SYMMOL, respectively, which either could not be determined by the program or it was falsely assigned. These values are much higher in case of NWChem, ORCA and MRCC, being 98, 59 and 166, respectively. Table 2 shows the ratio of correctly identified point groups to those which could be detected. The stand-alone software all perform well (above 90% accuracy), though the performance of the other three are much worse, interestingly. ORCA was able to generate the correct point group in 176 cases, but it was unable to supply one in 52. NWChem was much less accurate (137 correct point groups, 68.5% of all determined), and it could not give a point group for 35 geometries. In 42 cases MRCC could only determine a subgroup of the full point group of the molecules. Surely, these performances could be improved by further variations of the tolerances used in the calculations, yet this could not be achieved in a reasonable time. In the end, we can say that SYVA is the most efficient in the determination of symmetry of the seven programs examined.

Symmetry, Symmetrizer and SYVA list the found symmetry elements, and Symmetrizer, SYMMOL and SYVA can also symmetrize the structures. Furthermore, SYVA can generate the framework group of the molecule, give the symmetries of the normal vibrational modes and find the highest symmetry within a certain tolerance automatically. The other programs do not have such capabilities.

Br	0.00000	0.00000	0.00000	N	2.0	0.0	0.0
F	-1.85720	0.00000	0.00000	N	0.0	2.0	0.0
F	1.85720	0.00000	0.00000	N	-2.0	0.0	0.0
F	0.01850	-0.00210	1.85880	N	0.0	-2.0	0.0
F	0.01680	1.85560	-0.00070	S	3.0	-3.0	3.0
F	0.01580	-0.00070	-1.85840	S	-3.0	3.0	3.0
				S	3.0	3.0	-3.0
				S	-3.0	-3.0	-3.0
(a) Bromine pentafluoride.				(b) N ₄ S ₄ .			
C	0.00000	0.00000	2.84310	C	1.6729	-0.6929	2.0034
C	0.00000	0.00000	-2.84310	C	1.6729	0.6930	2.0034
C	-0.19180	1.44490	0.77770	C	0.6929	1.6729	2.0034
C	-0.19180	1.44490	-0.77770	C	-0.6930	1.6728	2.0034
C	0.00000	0.00000	-1.30330	C	-1.6729	0.6928	2.0034
C	1.34720	-0.55630	-0.77780	C	-1.6728	-0.6930	2.0034
C	1.34720	-0.55630	0.77770	C	-0.6928	-1.6730	2.0034
C	0.00000	0.00000	1.30330	C	0.6930	-1.6728	2.0034
C	-1.15540	-0.88850	0.77770	H	2.6436	-1.0948	2.0064
C	-1.15550	-0.88850	-0.77770	H	2.6434	1.0952	2.0064
H	0.12410	-0.93440	3.17690	H	1.0948	2.6435	2.0064
H	-0.87130	0.35980	3.17690	H	-1.0950	2.6433	2.0064
H	0.74730	0.57460	3.17690	H	-2.6434	1.0948	2.0064
H	0.12400	-0.93430	-3.17690	H	-2.6434	-1.0951	2.0064
H	0.74730	0.57470	-3.17690	H	-1.0948	2.6435	2.0064
H	-0.87140	0.36000	-3.17690	H	1.0952	-2.6434	2.0064
H	0.55570	2.02050	1.10950	Th	0.0000	0.0000	0.0000
H	-1.06350	1.80560	1.10950	C	-1.6729	0.6929	-2.0034
H	-1.06350	1.80570	-1.10940	C	-1.6729	-0.6930	-2.0034
H	0.55560	2.02050	-1.10950	C	-0.6929	-1.6729	-2.0034
H	2.09540	0.01830	-1.10950	C	0.6930	-1.6728	-2.0034
H	1.47200	-1.49130	-1.10950	C	1.6729	-0.6928	-2.0034
H	2.09540	0.01820	1.10940	C	1.6728	0.6930	-2.0034
H	1.47210	-1.49140	1.10950	C	0.6928	1.6730	-2.0034
H	-2.02760	-0.52910	1.10950	C	-0.6930	1.6728	-2.0034
H	-1.03200	-1.82380	1.10950	H	-2.6436	1.0948	-2.0064
H	-1.03190	-1.82370	-1.10940	H	-2.6434	-1.0952	-2.0064
H	-2.02760	-0.52900	-1.10940	H	-1.0948	-2.6435	-2.0064
(c) 1,4-dimethyl-bicyclo[2.2.2]octane.				H	1.0950	-2.6433	-2.0064
				H	2.6434	-1.0948	-2.0064
				H	2.6434	1.0951	-2.0064
				H	1.0948	2.6436	-2.0064
				H	-1.0952	2.6434	-2.0064
(d) Bis(cot)thorium(IV).							
O	0.72570	-2.23280	0.05820	C	0.0000	0.0000	-0.0000
O	-0.72360	2.23280	0.05690	C	0.0000	1.2358	0.9393
N	0.67330	0.00000	0.00040	H	0.8768	1.2585	1.5934
N	2.69590	-1.20070	0.06520	H	-0.8769	1.2585	1.5935
N	-0.67330	0.00000	-0.00050	Br	0.0000	2.9551	0.0227
N	-2.69480	1.20290	0.06530	C	-1.2358	-0.0000	-0.9393
C	1.33600	-1.19310	0.04170	H	-1.2586	0.8769	-1.5934
C	-1.33490	1.19360	0.04090	Br	-2.9551	0.0000	-0.0227
H	1.17270	0.87840	-0.00400	H	-1.2586	-0.8768	-1.5934
H	3.13400	-2.11000	0.09920	C	1.2359	0.0000	-0.9393
H	3.20120	-0.32620	0.04730	Br	2.9551	0.0000	-0.0226
H	-1.17310	-0.87820	-0.00530	H	1.2586	0.8769	-1.5933
H	-3.13170	2.11290	0.09920	H	1.2586	-0.8768	-1.5934
H	-3.20130	0.32900	0.04830	C	0.0000	-1.2359	0.9393
(e) Diurea.				H	0.8769	-1.2586	1.5934
				Br	-0.0000	-2.9551	0.0226
				H	-0.8769	-1.2586	1.5933
				(f) Tetrabromoneopentane.			

Fig. 4. Some example input geometries. For the results obtained from these structures see Table 3.

Table 2

Percentages of correctly determined point groups by various programs.

Program	SYVA	Symmetry	Symmetrizer	SYMMOL	NWChem	ORCA	MRCC
Accuracy ^a	99.6%	97.5%	93.2%	95.0%	68.5%	96.2%	82.1%
Unknown ^b	0	0	0	15	35	52	0

^a Based on the determined point groups only.^b The number of point groups the programs could not identify.

SYVA can also be integrated with other programs. Some examples are GAMESS [43,44] and MRCC [42], which are capable

of utilizing molecular symmetry. However GAMESS cannot determine the molecular point group and MRCC does not always

Table 3

Point groups assigned to some molecules by various programs. For the used input geometries, see Figs. 3 and 4.

Molecule	SYVA	SYMMOL	Symmetry	Symmetrizer	NWChem	ORCA	MRCC
Bromine pentafluoride	C _{4v}	C _{4v}	C _s	C _{4v}	– ^a	C _{4v}	C ₂
Diurea	C _{2h}	C _{2h}	C ₂	C _{2h}	C _{2h}	C _{2h}	C ₂
1,4-dimethyl-bicyclo[2.2.2]octane	D _{3h}	C _{3h}	D _{3h}	D _{3h}	C _{3h}	D _{3h}	D _{3h}
Bis(cot)thorium(IV)	D _{8h}	D _{8h}	D _{8h}	D _{4h}	– ^a	– ^a	D _{8h}
N ₄ S ₄	D _{2d}	D _{2d}	D _{2d}	D _{2d}	D _{2d}	D ₂	D _{2d}
Tetrabromoneopentane	D _{2d}	D _{2d}	D _{2d}	D _{2d}	S ₄	D ₂	D _{2d}

^a The program could not determine the molecular point group.

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*****
Program SYMMETRY determines all the symmetry elements
and symmetry operations of rigid molecules.
(C) G. Tasi, L. Gyevi-Nagy, R. Tobias, T.S. Tasi
Department of Applied and Environmental Chemistry
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H-6720 Szeged
Hungary
*****

-- Delta (distortion parameter)= 0.10000000

-- Title: Tetracyclopropylmethane
-- Molecular geometry (Angstroms) --

Atomic
number      x          y          z          w
6      -0.097600    1.511710   -1.314340   12.000000
6      -1.087720    1.645610   -2.538920   12.000000
6      -2.605390    1.879630   -2.388410   12.000000
6      -2.088540    0.519400   -2.871830   12.000000
6      1.031240     0.548780   -1.859830   12.000000
6      2.500260     0.599160   -1.390850   12.000000
6      2.134560     1.077030   -2.801290   12.000000
6      0.470830     2.907780   -0.833910   12.000000
6      -0.453730    4.139660   -0.725260   12.000000
6      0.669710     4.120390   -1.768210   12.000000
6      -0.802810     0.950230   -0.015640   12.000000
6      -0.955030    -0.554330    0.294340   12.000000
6      0.020390     0.295730    1.114560   12.000000
1      -3.037740    1.878400   -1.389310    1.007830
1      -3.072730    2.502000   -3.153270    1.007830
1      -2.258780    0.315040   -3.930710    1.007830
1      -2.205580    -0.295360   -2.160140    1.007830
1      -0.670730    2.066800   -3.454440    1.007830
1      0.708460     -0.445970   -2.173290    1.007830
1      2.800100     1.385260   -0.700870    1.007830
1      3.021090     -0.358710   -1.342150    1.007830
1      2.211020     2.152150   -2.955260    1.007830
1      2.447950     0.411660   -3.606960    1.007830
1      1.174690     2.883830    0.001560    1.007830
1      -1.459240     4.070170   -1.136110    1.007830
1      -0.281660     4.805480    0.121940    1.007830
1      1.524110     4.762770   -1.545740    1.007830
1      0.339450     4.040830   -2.801700    1.007830
1      -1.605320     1.562780    0.404430    1.007830
1      -1.871580     -0.846950   -0.810370    1.007830
1      -0.489340    -1.273360   -0.377390    1.007830
1      -0.300700     0.503710    2.136940    1.007830
1      1.073640     0.088720    0.932950    1.007830

-- SYMMETRY OPERATIONS --

# 1: COM -- with atom C (# 1)
# 2: C (2) -- with 1 unmoved atoms
# 3: S (4) -- with 1 unmoved atoms
# 4: S (4^3)

      g      E      i      SG      Cn      Sn
-----
      4      1      0      0      1      2

Distorsion of the found symmetry elements: 0.00000000
Distorsion of geometry due to symmetry elements: 0.05290492

-- POINT GROUP --

-- The structure should belong to the S4 point group.

S4 = {(E) (C2) (S4)}

      g      E      i      SG      Cn      Sn
-----
      4      1      0      0      1      2

The molecule is not polar and not chiral.

-- FRAMEWORK GROUP --

Framework group: S4[0(C),X(C12,H20)]

-- Rotational symmetry number: 2

-- VIBRATIONS --

Irreps of the vibrational normal coordinates:

23 A + 24 B + 23 E

There are 70 different vibrational frequencies:

- 47 IR active vibrations, with symmetry: B, E
- 70 Raman active vibrations, with symmetry: A, B, E

-- VARYING TOLERANCE --

Scanning range: 0.00500000 < delta < 0.10000000

S4 point group found in range 0.05290492 < delta < 0.10000000
C2 point group found in range 0.01007113 < delta < 0.05184682
C1 point group found in range 0.00146040 < delta < 0.00986971

Largest symmetry should be S4.

Available subgroups:
C1, C2, S4

-- OPTIMIZING S4 SYMMETRY

-- OPTIMIZING SYMMETRY ELEMENTS --

Nothing to optimize.

-- OPTIMIZING GEOMETRY --

Number of optimized atoms: 33

Geometry optimization successful.

```

Fig. 5. The results of a calculation performed on tetracyclopropylmethane with distorted geometry.

find the highest symmetry. It would be beneficial to use these programs with the symmetry provided by SYVA to reduce the time of computations.

After some initializations the subroutines of SYVA may be called to detect the maximal order point group of the given geometry (var_delta subroutine). Knowing the largest symmetry it is straightforward to extract the symmetry unique atoms (one of each class handed by sym_elements subroutine) and orient the molecule according to the master frame used by GAMESS with the help of the transformations implemented in SYVA. MRCC performs this transformation itself after the symmetry analysis.

This way users may avoid, for example, the necessity of point group determination and orientation of the geometry separately from the GAMESS software, either by themselves or with an other program. Only the full geometry needs to be specified to get these structural properties. It can be seen from Table 4 that the use of the point group symmetry from SYVA significantly reduces the computation complexity.

The two additional files GAMESS and MRCC supplied with SYVA make it easier to integrate it with the aforementioned programs. Further help with the modifications is available upon request to the authors.

Table 4

Running times of GAMESS (RHF/cc-pVDZ energy calculation) on various molecules using no symmetry and the point group supplied by SYVA.

Molecule	Computation time(s)		Point group ^c
	C ₁ ^a	DET ^b	
Pentane	17.425	5.311	C _{2v}
Benzene	10.495	0.744	D _{6h}
N ₄ S ₄	10.508	3.794	D _{2d}

^a Ignoring symmetry (specifying C₁ point group).

^b Using detected point group (last column).

^c Determined by SYVA.

```

*****
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H-6720 Szeged
Hungary
*****

-- Delta (distortion parameter)=      0.00100000

-- Title: TCPM

-- Molecular geometry (Angstroms) --

Atomic
number      x      y      z      w
6      0.000000      0.000000      0.000000      12.000000
6      -0.990453      0.134823      -1.223498      12.000000
6      -2.508123      0.368843      -1.072988      12.000000
6      -1.991273      -0.991387      -1.556408      12.000000
6      1.128534      -0.962171      -0.544758      12.000000
6      2.597947      -0.907045      -0.877289      12.000000
6      2.229012      -0.433093      -1.488073      12.000000
6      0.569992      1.394872      0.474938      12.000000
6      -0.356635      2.621167      0.610507      12.000000
6      0.741932      2.610896      -0.459119      12.000000
6      -0.708073      -0.567523      1.293319      12.000000
6      -0.831755      -2.072694      1.609396      12.000000
6      0.118896      -1.196687      2.433974      12.000000
1      -2.940473      0.367613      -0.073888      1.007830
1      -2.975463      0.991213      -1.837848      1.007830
1      -2.161513      -1.195747      -2.615288      1.007830
1      -2.108313      -1.806147      -0.844718      1.007830
1      -0.573463      0.556013      -2.139018      1.007830
1      0.807290      -1.957051      -0.855362      1.007830
1      2.898691      -0.117263      0.608951      1.007830
1      3.121021      -1.863362      -0.026159      1.007830
1      2.302755      0.641093      -1.645302      1.007830
1      2.544744      -1.100487      -2.292355      1.007830
1      1.289690      1.369558      1.293993      1.007830
1      -1.371982      2.546988      0.226316      1.007830
1      -0.166391      3.285563      1.455703      1.007830
1      1.599712      3.254075      -0.255907      1.007830
1      0.384265      2.536759      -1.484647      1.007830
1      -1.523517      0.031480      1.700387      1.007830
1      -1.745270      -2.381926      2.119913      1.007830
1      -0.342483      -2.787109      0.949584      1.007830
1      -0.216839      -0.989329      3.451940      1.007830
1      1.177540      -1.381934      2.263705      1.007830

Distorsion of the found symmetry elements:      0.00000000
Distorsion of geometry due to symmetry elements: 0.00000007

-- POINT GROUP --

-- The structure should belong to the S4 point group.

-- VARYING TOLERANCE --

S4 point group found in range 0.00000007 < delta < 0.05000000

Largest symmetry should be S4.

```

Fig. 6. The results of a calculation performed on tetracyclopropylmethane with symmetrized geometry (S₄).

6. Summary

Molecular symmetry can be used in many cases in calculations. In the present article, a simple vector algebraic method is presented which is capable of determining every symmetry element of an arbitrary molecule. The described method is implemented in Fortran language. The program is suitable, in addition to determining symmetry elements and the point group, for generating the framework group suggested by John Pople, performing some spectroscopic calculations and optimizing the geometry according to a certain symmetry. It is also possible to generate input files to

other programs (e.g. Gaussian) to perform further calculations on the symmetrized geometries. The performance of the program was demonstrated by comparison of running times and correctly determined point groups with other widely used stand-alone (SYMMOL, Symmetrizer), as well as molecular modeling (NWChem, ORCA, MRCC) software. Out of the programs examined the performance of SYVA was the best on the test set. A possible application was also illustrated as integration with other programs (GAMESS and MRCC). The program is distributed as source code under the GNU Lesser General Public License, which allows users to modify the code for their own requirements or link it with other software.

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